

MARC NEU

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PROFESSIONAL SUMMARY

Hardware and machine learning co-design researcher specializing in deploying point cloud networks on FPGAs and heterogeneous SoCs under hard real-time constraints. Currently leading the development of an MLIR-based compiler that lowers quantized ONNX models to HLS and AMD AI Engine targets. Designed the first dynamic point cloud network trigger in operation at a particle physics experiment, running with an end-to-end latency of 1.067 μ s. Brings research prototypes into production, covering algorithm mapping, RTL implementation, verification, and on-site commissioning.

PROFESSIONAL EXPERIENCE

Visiting Researcher

Oct – Dec 2026

High Energy Accelerator Research Organization (KEK), Tsukuba, Japan

- **Upgrade and commissioning:** Upgrading the GNN-ETM trigger on site within the Belle II data acquisition system and validating it on recorded data for the 2027 run period.

Researcher & Ph.D. Candidate

2022 – 2026

Institute for Information Processing Technologies (ITIV), Karlsruhe Institute of Technology (KIT)

- **Dataflow compiler:** Leading the ongoing development of DeePloy, an MLIR-based compiler that lowers quantized ONNX models to Vitis HLS C++ and AMD AI Engine kernels through custom dialects and quantization-aware passes.
- **Machine learning hardware trigger:** Designed and commissioned the GNN-ETM, an FPGA-based graph neural network trigger for the Belle II calorimeter. In operation since 2025 with an end-to-end latency of 1.067 μ s.
- **Acceleration on heterogeneous SoCs:** Mapped a dynamic point cloud network onto AMD Versal VCK190 by partitioning it across FPGA and AI Engines, fusing operators, and optimizing kernels. Achieved 7.02 $\times$ the throughput of an NVIDIA L40S GPU (IEEE FCCM 2026 Computing Challenge).
- **Joint communication and sensing DAQ:** Built a massive MIMO OFDM data acquisition platform with 32 RFSocS, supporting up to 256 channels at 1 GHz baseband bandwidth each.
- **Teaching and supervision:** Taught the System-on-Chip Laboratory course and supervised 14 bachelor's and master's theses.

Research Assistant

2020 – 2021

Karlsruhe Institute of Technology (KIT), Karlsruhe

- Contributed to the design and verification of an SSD-based PCIe data acquisition system for the Tandem-L radar satellite mission.

Working Student in Application Engineering

2019 – 2020

Infineon Technologies AG, Munich

- Performed PSpice circuit simulations and evaluated interface concepts between IGBT gate drivers and power management ICs.

Intern in Hardware and Logic Engineering

2019

Airbus Defence and Space, Ottobrunn

- Designed the digital controller for a proof-of-concept atomic clock as part of the bachelor's thesis.

EDUCATION

Ph.D. Electrical Engineering and Information Technology <i>Summa cum laude, Karlsruhe Institute of Technology (KIT), Karlsruhe</i> Thesis: Template-based Deployment for Real-Time Deep Learning on Point Clouds	2022 – 2026
M.Sc. Electrical Engineering and Information Technology <i>Grade 1.1 (with distinction), Karlsruhe Institute of Technology (KIT), Karlsruhe</i>	2019 – 2022
B.Sc. Electrical Engineering and Information Technology <i>Grade 1.7 (good), Karlsruhe Institute of Technology (KIT), Karlsruhe</i>	2016 – 2019

SELECTED PUBLICATIONS

- M. Neu**, F. Baptist, T. Lobmaier, F. Papagno, T. Ferber, J. Becker, "Reconfigurable Computing Challenge: Real-Time Graph Neural Networks for Online Event Selection in Big Science," in *Proc. IEEE FCCM*, Atlanta, USA, 2026.
- I. Haide, **M. Neu**, Y. Unno et al., "Real-time graph neural networks on FPGAs for the Belle II electromagnetic calorimeter," submitted to JINST, 2026.
- M. Neu**, I. Haide, T. Justinger, T. Rädler, V. Dajaku, T. Ferber, J. Becker, "Real-Time Graph-based Point Cloud Networks on FPGAs via Stall-Free Deep Pipelining," in *Proc. IEEE SBCCI*, Manaus, Brazil, 2025.
- M. Neu**, J. Becker, P. Dorwarth et al., "Real-Time Graph Building on FPGAs for Machine Learning Trigger Applications in Particle Physics," *Computing and Software for Big Science*, vol. 8, no. 8, 2024.
- M. Neu**, C. Karle, B. Nuss, P. Groeschl, J. Becker, "A Scalable and Cost-efficient Antenna Testbed using FPGA-Server Compound Structures for Prototyping 6G Applications," in *Proc. IEEE DCOSS-IoT*, Coral Bay, Cyprus, 2023.

TECHNICAL SKILLS

Hardware Design

SystemVerilog, Chisel, Vitis HLS, AMD Versal AI Engines, AMD RFSocS, IP cores (GT transceivers, RF data converters, 100G CMAC), digital signal processing

Compilers & Machine Learning Deployment

MLIR, LLVM, CIRCT, onnx-mlir, QONNX, graph neural networks, point cloud networks

Programming

C++, Python, Scala, Tcl

Tools & Verification

AMD Vivado, AMD Vitis, QuestaSim, cocotb, CMake, Git, Linux

AWARDS & HONORS

Leadership Talent Academy Scholarship <i>Karl Schlecht Foundation</i>	2021
Deutschlandstipendium (Germany Scholarship) <i>Federal Ministry of Research, Technology and Space</i>	2021

LANGUAGES

German (native), English (fluent)